

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Social Network Analysis [NPT_4146]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider the following code snippet and answer the questions below:

A)

```
import networkx as nx
G = nx.Graph()
G.add_nodes_from(["Alice", "Bob", "Charlie", "David", "Eve"])
G.add_edges_from([("Alice", "Bob"), ("Alice", "Charlie"), ("Bob", "David"), ("Charlie", "Eve")])
G.remove_edge("Alice", "Charlie")
G.remove_node("David")
has_edge = G.has_edge("Alice", "Charlie")
neighbors_of_alice = list(G.neighbors("Alice"))
```

 (5)

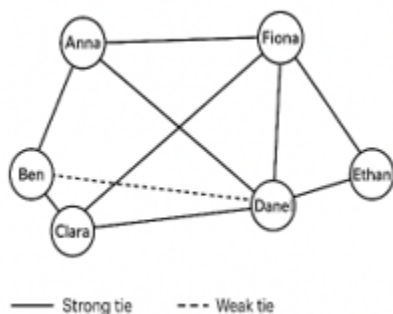
Apply NetworkX graph manipulation functions such as `remove_edge()`, `remove_node()`, `has_edge()`, and `neighbors()` to modify and analyze the given social network, interpret how these operations affect relationships among individuals, and extend the network by adding a new connection between Bob and Eve.

- B) **Apply** graph metrics such as **density**, **average degree**, and **clustering coefficient** in **Gephi** to analyze the overall structure of a network, and interpret how these metrics are computed and visually represented within the software. (3)

- C) **Apply** the concept of **closeness centrality** to evaluate its **limitations** in large, real-world networks such as **social media platforms**, and propose ways to **address computational challenges** while identifying an **alternative centrality measure** that offers greater efficiency for large-scale analysis. (2)

- 2) (5)

A)



Examine the professional network of six individuals-**Anna, Ben, Clara, Daniel, Ethan, and Fiona**-based on the nature and frequency of their interactions. Identify and distinguish between **strong** and **weak ties** within the network, and analyze how the combination of these ties influences **information flow, collaboration, and cohesion** in the overall professional community.

- B) **Analyze** the social network consisting of four individuals - **Aarav, Meera, Rohan, and Priya** - where the relationships are defined as follows:

- Aarav is connected to Meera and Rohan.
- Meera is connected to Rohan and Priya.
- Rohan is connected to Priya.

(3)

Based on these connections, **identify all possible triads** within the network and **distinguish** whether each triad is **open** or **closed**, explaining how these structural patterns reflect the **cohesion and balance** of the network.

- C) In a **professional networking platform of 800 users**, a **popular industry leader** shares a new online course. In the first round, the leader influences **8 people** to enroll. Each of those influenced individuals then convinces **4 more people** in the next round. **Apply** the concept of **information diffusion** to calculate how many users have enrolled in total after **two rounds** of influence spread. (2)

- 3) **Apply** the concept of **spatial segregation** to examine how it manifests differently in **physical social networks** (such as urban neighborhoods) and **online social networks**, identifying the mechanisms driving segregation in each and the resulting **social and economic impacts**. (5)

- A)
- B) **Analyze** the given social network triad involving **Alice, Bob, and Carol** using **balance theory** to determine whether the triad is **balanced** or **unbalanced**, and justify your reasoning by examining the pattern of positive and negative relationships among them. The table below represents the relationships among **three individuals in a social network triad**:

Relationship	Type	Symbol
Alice – Bob	Positive	(+)
Alice – Carol	Positive	(+)
Bob – Carol	Negative	(-)

(3)

- C) **Apply** the concept of **network centrality** to explain how increasing **positive relationships** around a **highly central node** can contribute to the **stability and cohesion** of the overall social network. (2)

- 4) Analyze how the presence of highly connected hubs in a **scale-free network** affects the **efficiency and fairness** of information dissemination across a digital platform. Assess both the **advantages** of rapid diffusion through central hubs and the **risks** of network vulnerability or misinformation concentration, and **propose** strategies to enhance network robustness while maintaining effective communication flow. (5)

- A)

- B) Compare the **Watts-Strogatz (WS) model** and the **Erdős-Rényi (ER) model** in terms (3)

of how they represent **network connectivity and clustering**, and **illustrate** how the process of **rewiring edges** in the WS model leads to the emergence of **small-world properties** that differ from the random connections in the ER model.

- C) Apply the concept of **network structure** to **analyze** how patterns of connectivity influence the **spread of an epidemic** across a population. **Explain** why networks containing **high-centrality** or **hub nodes** are more **susceptible to rapid disease transmission**, and **interpret** how variations in network topology affect the overall **speed and reach** of the outbreak. (2)
- 5) Apply the **mathematical principles of the small-world model to demonstrate** how short path lengths and high clustering coefficients contribute to efficient connectivity in networks. (5)
- A) **Analyze** how this model helps **interpret** real-world phenomena such as **the six degrees of separation, viral marketing, and information diffusion** across social systems.
- B) **Apply** and **illustrate** how the **cascading model** explains the **progressive spread of information** within a social network, where each influenced user can further impact others. (3)
- C) Apply the concept of **network diameter and average path length to analyze** how the overall connectivity of a social network influences the **efficiency of communication and the speed of information spread**. **Interpret** how shorter path lengths contribute to the **formation of tightly connected communities** and enhance the **potential for viral diffusion** within the network. (2)

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Event Name	Subject
058-END SEMSTER-ODD-2025	NPT_4146:Social Network Analysis

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	NPT_4146:Social Network Analysis

#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	31(62%)
4	Analysis	5	19(38%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)